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May 24, 2026

The Polynomial Ring $\mathbb{Z}[x]$ (Part One)

A term that you should become familiar with is **ostensive definition**. To define a word ostensively, you give examples of things that would be classified as whatever the word is. There are also **formal definitions** in mathematics, but they are often extremely difficult to comprehend without sufficient experience. For this reason, I will resort to using ostensive definitions as much as possible. Note, however, that deep mathematics requires formal definitions. Ostensive definitions are only for pedagogical purposes.

Let me ostensively define what **polynomials** are. They are things like these:

$$\begin{aligned}&x^2+2x+3,\\&x^4-6x^3+9x-1,\\&x^{11}-5x,\\&x^{27}-8x^{14}+9x^3-x\end{aligned}$$

The numbers in front of the powers of x are called **coefficients**. For example, the coefficient on the x^3 term in the second polynomial above is -6. Knowing this word is crucial for fluently conversing in the language of mathematics. Do not worry about what x is. It is unknown, and we are **<u>not</u>** trying to solve for it.

Take note of the following: Polynomials themselves are **<u>not</u>** equations. There is such a thing as a polynomial equation, but the entities above have no equals signs in them. Thus, they are not equations. To further clarify what polynomials are, let me provide you with some **non-polynomials** (things that aren't polynomials):

$$\begin{aligned}&e^{2x}+7,\\&\ln(5x+2)+\sqrt{x},\\&e^x+x^6+2x-1,\\&6\cos(x)+8\sin(x)\end{aligned}$$

If you take all the polynomials with coefficients in $\mathbb{Z}$, you get a ring that we denote by $\mathbb{Z}[x]$. You can call it **the ring of polynomials with coefficients in** $\mathbb{Z}$. Remember: a ring is a set equipped with **<u>two</u>** operations. For today's lesson, we will only concern ourselves with the addition operation. We will save multiplication for next time. Let me ostensively define polynomial addition. Let's add together the polynomials $4x^3-5x^2+x-7$ and $-9x^7+5x^3+12x^2+9$. I'm going to start by putting each of these in parentheses for the sake of readability and distinction. We have

$$\begin{aligned}
(4x^3 - 5x^2 + x - 7) + (-9x^7 + 5x^3 + 12x^2 + 9) &= -9x^7 + 4x^3 + 5x^3 - 5x^2 + 12x^2 + x - 7 + 9 \\
&= -9x^7 + (4x^3 + 5x^3) + (-5x^2 + 12x^2) + x + (-7 + 9) \\
&= -9x^7 + 9x^3 + 7x^2 + x + 2.
\end{aligned}$$

Look at the above calculation very carefully. To perform polynomial addition, you simply add coefficients on corresponding powers of x together. If one of your polynomials had $6x^3$ in it and the the other had $7x^3$ in it, your resulting polynomial would have $13x^3$ in it. Note that terms like $3x^8$ and $2x^2$ cannot be added together, because their powers of x are not the same.

Exercises

(1.) Find the sum of $x^4 + 9x^2$ and $x^5 - 2x^4 + 2x^2$.

(2.) Find the difference of $2x^3 + 4x^2$ and $7x^3 + 2x^2$.

(3.) Prove (informally) that $\mathbb{Z}[x]$ forms an Abelian group under polynomial addition ("Abelian" means "commutative").